Automatic Race Timing

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Introduction: Modernising Race Timing Systems

- Running races across abilities, ages, and forms require accurate timing solutions.
- **Current methods:**
  - RFID: Effective but expensive and complex for low-budget races.
  - Manual recording (clipboard + stopwatch): Cost-effective but prone to human error and inefficiencies (e.g., during high participant volumes).
- **Proposed solution:**
  - Leverages computer vision and readily available technologies (cameras and laptops).
  - Automates race timing by recognising bib numbers and detecting finishes via camera feeds.
  - Results are uploaded directly to the organiser's website.

Project Justification: Why Our Solution is Superior

- **Cost-Effective Alternative:** Utilises widely available tech (cameras, laptops) at a lower cost than RFID systems.
- **Increased Accessibility:** Avoids the technical challenges and organisational demands of RFID, empowering organisers of all skill levels.
- **Enhanced Accuracy:** Comparable to RFID, eliminating human error associated with manual methods.
- **Improved Scalability:** Handles high volumes of participants with ease, offering a practical solution for larger races.
- Harnesses the potential of modern technologies to make race timing smarter, cheaper, and more efficient for all.

Overall System Architecture

- **Modular Microservice Approach:** The system adopts a modular architecture, with separate microservices for the computer vision (CV) system and the website.
- **Separation of Concerns:** Each component performs a specific role, ensuring independent development, testing, and scalability.
- **Inter-service Communication:** Communication between the CV module and the website is achieved via secure RESTful APIs, enabling seamless data transfer and reducing integration complexity.
- **Scalability and Robustness:** The decoupled design ensures that each component can be independently scaled and updated without affecting the other.

Structure of CV

- **Why Separate the CV System:**
  - Reduces networking load by locally running detection and OCR instead of live streaming the video to a back-end server.
  - Allows for flexible deployment of the CV model on local hardware.
  - The CV module can potentially be integrated into other race timing systems or projects, promoting reusability.
- **CV System Features:**
  - Takes camera input to capture live video of the race finish line.
  - Uses advanced computer vision models to:
    - Detect runner bibs.
    - Extract bib numbers using OCR.
    - Detect when participants cross the finish line by detecting when the runner leaves the view of the camera.
    - Reptimestamps for which the that runner crosses the finish line to the backend server.

Structure of Website

- **Why the Website is Designed This Way:**
  - Directly addresses client requirements for usability, accessibility, and functionality.
  - Provides an intuitive interface for managing races, participants, and results.
  - Ensures compatibility with the CV system, allowing seamless data upload and processing.
- **Website Features:**
  - **User Management:** User login, registration, and password reset.
  - **Race Management:** Create races and upload participant details.
  - **Results Management:** Upload race results, automatically sort, and filter leaderboards.
  - **Scalability and Optimisation:** Designed to handle large numbers of users and races, with optimisation for quick and efficient data retrieval.

Implementation Overview: Project Decomposition

The project can be broken down into the following modular components:

- Frontend
- Backend
- Computer Vision Model

Frontend Walkthrough

- **User Login/Registration:** The web interface should allow users to register accounts, allowing for authentication and authorisation
- **Race Creation:** The frontend should allow race organisers to configure their own races, as well as configure who is participating in their races.
- **Data Upload:** Organisers and regular users are able to view the results of the race live, as they come in, processing them so that they can be filtered and sorted

Backend Walkthrough

- **Website database interactions:** Using RESTful API requests the Django backend interacts with the SQLite database through SQL statements
- **Result upload from CV models:** Through requests containing CSV files, the backend allows for processing of results from the Computer Vision model script
- **Result upload to the Web Interface:** After processing the results delivered by the locally run CV script results are be uploaded to the frontend web interface so that they can be presented

Computer Vision Model Walkthrough

- **Camera Input:** In order to process the video live, the laptop is able to take in video input from a camera near the finish line
- **Bib Detection/OCR:** Runners are be recognised and their bibs are located and bib numbers extracted
- **Timing Calculation:** The finish time for each runner is calculated and sent to the backend.

Operational Setup: Overview

- **System Overview:**
  - The system consists of three key components: the computer vision (CV) model, the camera setup, and the website.
  - Designed to operate seamlessly using widely available hardware and standard hosting options, ensuring ease of deployment.
- **Flexibility in Deployment:**
  - Modular design enables flexibility: components can be upgraded or replaced independently.
  - Scalable to support races of varying sizes, from small local events to larger competitions depending on hardware investment.

Operational Setup: The CV Model

- Runs efficiently on common laptops, minimising hardware requirements via hardware acceleration for lightweight and fast processing.
- Designed to process live camera feeds in real time, identifying runners and detecting bib numbers.
- Compatible with smartphones as cameras, requiring only a basic connection to transfer video to the laptop.

Operational Setup: The Website

- **Current Status:**
  - Currently runs on localhost for testing and development purposes.
- **Future Hosting Options:**
  - Transition to university hosting for wider accessibility and reliability.
  - Use affordable, scalable hosting options to support client needs without excessive costs.

Demo

Enjoy this demonstration of the product.

Planned Work: Multi Camera Setup

- Implement a multi-camera setup to enhance bib detection and eliminate uncertainties caused by occlusions or poor angles.
- Improved accuracy in detecting runners crossing the finish line.
- Redundancy provided by multiple cameras ensures robustness in various environmental conditions.
- Optimised processing pipeline to efficiently handle data from multiple video sources.

Planned Work: Edge Cases and Runner Pictures

- **High Priority Goal:** Save race footage to address edge cases.
  - Allows organisers to review footage manually for scenarios where bib detection fails.
- **Medium Priority Goal:** Develop a user-friendly system for manual handling of unrecognised bibs and their runners.
  - Provides a fallback mechanism to ensure no participant is left out of the results.
  - Integrates seamlessly into the existing leaderboard management workflow.
- **Low Priority Goal:** Capture and store runner pictures as part of the system, enabling visual proof of results and enhancing the participant experience.

Planned Work: University Hosting

- Migrate the website to university-hosted servers to transition off localhost.
- Enables public access to the website for race organisers and participants.
- Demonstrates scalability and prepares the system for potential deployment beyond academic settings.

Future Outlook and Completion

- **Current Status:**
  - The system is largely operational with core features implemented.
  - Basic CV and website functionalities are functional can be enhanced before final deployment.
- **Future Goals:**
  - Fine-tune the CV models for higher accuracy and broader edge case handling.
  - Complete website migration and add more peripheral features.